

HAMZA IDRISSE MELIANI

Engineering Student – Mathematics & Systems Engineering

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PROFILE SUMMARY

Engineering student at **Ecole Hassania des Travaux Publics (EHTP)**, specializing in Mathematics & Systems Engineering. Passionate about **Artificial Intelligence, Data Science, and optimization** applied to real-world problems. Proficient in Python and C, with practical experience in algorithm design, data analysis, and embedded systems simulation. Active member of the Hassania IT Club. Seeking a **summer internship (June–July 2026)** in AI, Data Science, optimization, or systems engineering.

EDUCATION

Engineering Degree – Mathematics & Systems Engineering 2025 – 2028

Ecole Hassania des Travaux Publics (EHTP) – Casablanca, Morocco

- Coursework: Optimization & Decision Making, Advanced Algorithmics & Game Theory, Numerical Modeling & Scientific Computing, Fitting Statistical Models, Probability & Statistics, Automation & Signal Processing, Networks & Databases

DEUST – Mathematics, Computer Science & Physics (MIP) 2023 – 2025

Faculté des Sciences et Techniques de Fès (FST) – Fès, Morocco

- Scientific foundation: C & Python programming, SQL/MySQL databases, mathematics, physics, and data structures

Baccalaureate – Physical Sciences, French Option, Mention Bien 2023

Lycée Qualifiant Abdelkrim Erraiss – Fès, Morocco

PROJECTS

Nine Men's Morris – AI Strategy Game Engine 2026

EHTP – Advanced Algorithmics & Game Theory | C Programming

- Developed a complete Nine Men's Morris engine in C with full rule enforcement: placement, movement, mill detection, and capture phases
- Built an AI opponent using **Minimax algorithm with Alpha-Beta pruning** for efficient game-tree search and optimal move selection
- Designed a modular architecture (board representation, game-state machine, rule engine) and ASCII board rendering

Automatic Animal Food Dispenser – Embedded Systems Simulation 2025

EHTP – Classical Computing Project | Arduino / Wokwi

- Designed and simulated an autonomous IoT feeding system on Wokwi using Arduino UNO
- Integrated RTC DS1307 for time-based scheduling, 4x4 keypad input, and I2C LCD 2004 status display
- Programmed servo motor control (0°–180°) via state-machine firmware | wokwi.com/projects/449173652836253697

Graphical Calculator – Python GUI Application 2024

FST Fès – Python / Tkinter Project

- Built a fully functional GUI calculator with dynamic expression rendering and operator-precedence handling using Python & Tkinter

TECHNICAL SKILLS

Languages	Python, C, C++, SQL
AI & Data Science	NumPy, Pandas, Matplotlib, Jupyter Notebook, Data Analysis, Game-Tree Search (Minimax, Alpha-Beta)
Algorithms & Math	Data Structures, Numerical Methods, Optimization & Decision Making, Statistical Modeling
Embedded Systems	Arduino UNO, Wokwi Simulation, Microcontroller Programming, I2C, Servo Control
Tools	Git, GitHub, VS Code, CLI / Version Control, MATLAB, Microsoft Excel, Word, PowerPoint

CERTIFICATIONS

- Python for Data Science, AI & Development – IBM Nov 2025
- Getting Started with Git and GitHub – IBM Dec 2025

EXTRACURRICULAR & VOLUNTEERING

Events Cell Member – Hassania IT Club (H.I.T.C), EHTP Oct 2025 – Present

- Organized coding competitions and technical workshops for engineering students, managing logistics end-to-end
- Contacted and coordinated with external speakers and evaluation panels for tech events

LANGUAGES

- Arabic – Native
- French – B1
- English – B1+